Choosing the local witnesses simultaneously

Human Compression project: L044 continuation

All marks below are involutions. Retain every actual odd-degree head and global paired-tail hypothesis of *Local pairs and ambient witnesses*, with $H = G_{\geq 5}$ and $K = G_{>5}$.

Theorem 1 (A colour with no local pair)**.** *There is a colour $j \geq 5$ such that every actual H-orbit containing an ambient-eligible j-edge and intransitive under K has exactly one edge eligible for its full local j-omission. Its active labels form an interval starting at 5, and δ fixes it pointwise. Every H-orbit contains at most one ambient-eligible j-edge, and at least two distinct H-orbits contain one.*

Proof. If no such colour existed, for each tail colour one could choose a K-intransitive actual H-orbit containing one ambient witness and two edges eligible for the full local omission. These independent choices would give exactly the mixed assignment excluded by the complete acquired proof. Thus a colour exists for which no such orbit has a local pair.

Every K-intransitive witness orbit has at least one locally eligible edge, namely its ambient witness. Hence it has exactly one. Transitivity of H makes its local omission have exactly two orbits, joined by that edge. If any H-orbit had two ambient j-edges, then either it would already be a forbidden K-intransitive target, or the complete whole-K-orbit transport would carry both edges simultaneously into such a target. Both would remain ambient and therefore locally eligible. This contradiction proves the global at-most-one assertion.

The acquired active-label argument applies to every witness orbit. A K-intransitive one starts at 5 and has a genuine local unique crossing. The complete orbit-size and parity proof therefore makes δ fix each of these orbits pointwise. The two ambient edges required globally lie in two distinct actual H-orbits. No transport of the second edge, or extra local uniqueness for a K-transitive orbit, is asserted here. □

Lemma 2 (Prefixes of an anchored action)**.** *Let $T = \langle t_0, \dots, t_{u-1} \rangle$ be transitive on V with string commutations. Suppose a is fixed by $t_1, \dots, t_{u-1}$ and $t_{u-1} \neq 1$. For $1 \leq v < u$, put $C_v = a^{\langle t_0, \dots, t_{v-1} \rangle}$ and $C_0 = \{a\}$. Then every inclusion $C_{v-1} \subset C_v$ is strict, so $|C_v| \geq v + 1$. Every mark of index at least $v + 1$ fixes C_v pointwise.*

Proof. A later mark commutes with each generator of the prefix and fixes a, proving the last assertion. If $C_v = C_{v-1}$ with $v < u$, then t_{v-1} preserves C_{v-1}; every mark of index at least v fixes C_{v-1}, by the same commutation argument. Thus C_{v-1} is invariant under all of T. Transitivity makes it V, on which t_{u-1} would act trivially, a contradiction. The strict inclusions and the degree bound follow. We make no strict-growth claim at $v = u$. □

Theorem 3 (A truncated local orbit)**.** *Let $Q = \langle \rho_m, \dots, \rho_y \rangle$ act transitively on W, with string commutations and all displayed restrictions nonidentity. Suppose its full local j-omission has exactly two orbits A, B joined by exactly one j-edge $\{a, b\}$. For $j + 1 \leq x \leq y$, at least $x - m + 1$ points of W are fixed by every mark of index greater than x.*

Proof. Set $L = Q_{<j}$ and $R = Q_{>j}$. On each of A, B the commuting product LR is transitive. Every mark at distance at least two from j fixes a, b, since it preserves the omission orbits and the unique crossing edge.

Put $v = x - j - 1 \geq 0$ and $R' = \langle \rho_{j+1}, \dots, \rho_{x-1} \rangle$, with $R' = 1$ if $v = 0$. The two sets

$$C_A = a^{LR'}, \qquad C_B = b^{LR'}$$

are disjoint and fixed pointwise by all marks after x: these marks fix both anchors and commute with every displayed generator of LR'. We bound their total size.

On each side the active labels of L form an interval ending at $j-1$, and those of R an interval starting at $j+1$, by the acquired commuting anchored-factor argument. Write their lengths as h_A, u_A and h_B, u_B. Then

$$\max(h_A, h_B) = j - m =: h, \qquad \max(u_A, u_B) = y - j.$$

All active left marks remain active on C_A or C_B as appropriate: their own orbit through the anchor is unchanged. The number of active right marks on these sets is respectively $\min(u_A, v)$ and $\min(u_B, v)$. If $v < u_A$, this follows from the strict prefix lemma; if $v \geq u_A$, it is the full right action. The same argument applies to B. Their maximum is v, since $v < y - j$.

Apply the complete two-commuting-factor size bound to the transitive actions on C_A, C_B. If the maxima h, v occur on opposite sides, the separate anchored bounds give total size at least $h + v + 2$. If they occur on one side, that side gives $h + v + 1$ and the other contributes one more. Empty factors are included.

The only possible exception would have $h = v = 1$, $|C_A| = 2$, $|C_B| = 1$, after interchanging sides. Then L has just one active mark, acting as the same transposition as ρ_{j+1} on C_A. Every further right mark fixes a and commutes with L, so fixes both points of C_A. It also fixes b, while the first right mark and L fix b already. Thus C_A, C_B are invariant under the complete LR, forcing $A = C_A$ and $B = C_B$. No right mark after $j+1$ could act nontrivially on W, contradicting $y \geq x = j + 2$. The exception is impossible. Hence

$$|C_A| + |C_B| \geq h + v + 2 = x - m + 1,$$

as required. No assertion about two disjoint point paths has entered the proof. □

Theorem 4 (Two witnesses with intransitive first omissions)**.** *For the colour of the first theorem, suppose two distinct witness orbits O_1, O_2 are both K-intransitive. Write their last active labels as $x \leq y$. Then $x < r - 1$. If $x = y$,*

$$|\operatorname{Fix}(G_{>x})| \geq 2x + 1.$$

If $j + 1 \leq x < y$, the proved bound is

$$|\operatorname{Fix}(G_{>x})| \geq 2x.$$

Proof. Both orbits are fixed pointwise by δ and have first active label 5. Their complete local orbit bounds give $|O_1| \geq x - 3$, $|O_2| \geq y - 3$. If $x = y$, both whole orbits and the seven head points are fixed after x, giving $2x + 1$. If $j + 1 \leq x < y$, the truncated theorem supplies at least $x - 4$ fixed points in O_2. Together with O_1 and the seven head points, this gives $(x-3) + (x-4) + 7 = 2x$.

If $x = r - 1$, then $y = r - 1$ too. The two complete orbit bounds and the seven head points occupy at least $2(r-4)+7 = 2r-1$ of the $2r+1$ points. The odd involution δ fixes this entire union pointwise, and must move at least two points outside it. Consequently exactly two points remain and δ is literally their transposition. The complete acquired exterior-pair obstruction applies under the present actual head and global ambient pair hypotheses. It excludes this possibility, so $x < r - 1$. □

Attribution and scope. The mixed-assignment contradiction, active intervals, whole-orbit transport, complete orbit-size bound and exterior-pair obstruction are acquired. The simultaneous witness choice and strict anchored-prefix count supply the new truncated bound. The unequal-end case currently provides $2x$, not the $2x+1$ fixed points required for the desired separator. The case $x = j < y$, a second witness with transitive K-action, and global attachments also remain. No complete L044 recovery, new induction, computation, formal or question credit is claimed.